Piecewise-Defined & Implicit Functions

Section 1: Evaluating Piecewise Functions

1 Let $f(x) = 2x + 1$ for $x < 2$, and $f(x) = x^2 - 3$ for $x \geq 2$. Find $f(2)$ and $f(-1)$.

a. $f(2) = 1$ and $f(-1) = -1$

b. $f(2) = 5$ and $f(-1) = -1$

c. $f(2) = 1$ and $f(-1) = -2$

d. $f(2) = 5$ and $f(-1) = 1$

2 A tax bracket levies no tax on the first 10,000 dollars, and 15% on income exceeding 10,000 dollars. Which piecewise function $T(x)$ models total tax for income $x \geq 0$?

a. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 0.15(x - 10000)$ for $x > 10000$

b. $T(x) = 0.15x$ for all x

c. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 0.15x$ for $x > 10000$

d. $T(x) = 0$ for $x \leq 10000$, and $T(x) = 1500 + 0.15x$ for $x > 10000$

3 Evaluate the ceiling function $g(x) = \lceil 2x - 0.3 \rceil$ at $x = 1.4$.

a. 3

b. 2

c. 2.5

d. 4

Section 2: Continuity of Piecewise Functions at Boundaries

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4 Find the value of k that makes $f(x)$ continuous at $x = 3$, where $f(x) = kx + 1$ for $x \leq 3$ and $f(x) = 2x^2 - 5$ for $x > 3$.

a. 4

b. 2

c. 5

d. 13

5 Let $g(x) = \frac{x^2 - 16}{x - 4}$ for $x \neq 4$, and $g(4) = c$. What value of c removes the discontinuity at $x = 4$?

a. 8

b. 4

c. 0

d. 16

6 A function h is defined by $h(x) = |x - 2| + 3$. Where is $h(x)$ continuous and where is it not differentiable?

a. Continuous everywhere; not differentiable at $x = 2$

b. Continuous only for $x \geq 2$; differentiable everywhere

c. Discontinuous at $x = 2$; differentiable elsewhere

d. Continuous and differentiable everywhere on $(-\infty, \infty)$

Section 3: Implicit Relations and Conic Sections

7 Identify the conic section described by the equation $4x^2 + 9y^2 - 16x + 54y + 61 = 0$.

a. Ellipse

b. Hyperbola

c. Parabola

d. Circle

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8 Find the center and radius of the circle given by $x^2 + y^2 - 6x + 8y = 0$.

a. Center $(3, -4)$, Radius 5

b. Center $(-3, 4)$, Radius 5

c. Center $(3, -4)$, Radius 25

d. Center $(6, -8)$, Radius 10

9 Which implicit relation fails the Vertical Line Test and therefore cannot be represented by a single function $y = f(x)$?

a. $x^2 + y^2 = 25$

b. $y = x^2 - 4$

c. $y = \sqrt{25 - x^2}$

d. $y = 3x + 1$

Section 4: Asymptotes and Branches of Hyperbolas

10 Find the equations of the asymptotes for the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

a. $y = \pm\frac{3}{4}x$

b. $y = \pm\frac{4}{3}x$

c. $y = \pm\frac{9}{16}x$

d. $y = \pm\frac{16}{9}x$

11 What is the orientation of the transverse axis for the hyperbola $\frac{(y-1)^2}{25} - \frac{(x+2)^2}{16} = 1$?

a. Vertical, along the line $x = -2$

b. Horizontal, along the line $y = 1$

c. Diagonal with slope $5/4$

d. Vertical, along the y-axis

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12 What are the vertices of the ellipse $\frac{x^2}{49} + \frac{y^2}{25} = 1$?

a. $(\pm 7, 0)$

b. $(0, \pm 5)$

c. $(\pm 5, 0)$

d. $(0, \pm 7)$

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